

Surname

Name

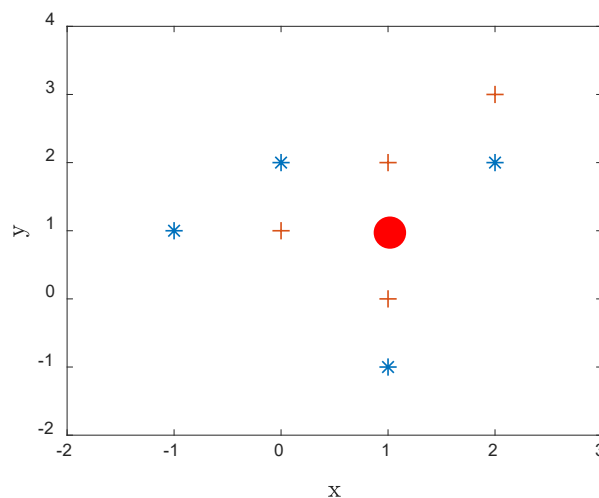
University ID Number

Signature

- =====
- Write the solutions (including procedures and intermediate steps) in the blank areas (use the back of the page, if needed)
 - The number of pages is 7. Additional papers will not be considered.
 - Use of books, lecture notes or any other material is forbidden. Electronics (smartphones, calculators, etc.) are forbidden.
 - Clarity, order and precision are strongly considered for the final evaluation.
- =====

1. [Nonparametric classification]

Suppose you are given a set of data where x and y are the 2 input variables and Class is the dependent variable, which can be either $*$ or $+$. The scatter plot below shows the data in the regressor space.



- 1.1. You want to predict the class of a new data point with coordinates $x = 1$ and $y = 1$ via the K-NN method using $K = 7$ and the Euclidian distance. Which class does this data point belong to?
- 1.2. Does the previous answer change if you use 3-NN instead of 7-KNN?
- 1.3. Which of the following statement is likely to be true as $\frac{1}{K}$ moves above 1?
 - A. The prediction error will increase;
 - B. The prediction error will decrease;
 - C. It does not make sense to have $\frac{1}{K} > 1$;
 - D. The prediction error will converge to a constant value.
- 1.4. Describe the K-NN prediction algorithm by also specifying under which assumptions it can be used.
- 1.5. Describe one practical tool to assess the prediction performance of a classifier.

Solution:

- 1.1 The new point will be classified as Class $*$, because there are 4 $*$ and 3 $+$ points in the nearest circle.
- 1.2 All 3 nearest point are of Class $+$, so this point will be classified as $+$.
- 1.3 The K-NN classifier can only contemplate integer values for the parameter K, thus it does not make sense to decrease K below 1. The only true statement is the third one.
- 1.4 See Chapter 4 of the lecture notes.
- 1.5 See Chapter 4 of the lecture notes.

2. [Fitting VS generalization]

Using the learning curves, describe the main trade-off between data fitting and model generalization properties.
Are such curves also related to the bias-variance trade-off?

Solution:

See [Chapter 2 of the lecture notes](#).

3. [Kalman filtering]

Consider the dynamical system

$$\begin{cases} x_1(t+1) = x_1(t) + v_{11}(t) \\ x_2(t+1) = x_1(t) + 2x_2(t) + v_{12}(t) \end{cases}$$

where v_{11} and v_{12} are white noises with zero mean, unitary variance and uncorrelated with each other. Let the initial condition be zero.

- 3.1 Are x_1 and x_2 stationary processes? Why?
- 3.2 Assume it is possible to measure only one state variable with a suitable sensor, affected by a disturbance $v_2(t)$, i.e., a white noise with zero mean, unitary variance, uncorrelated with v_{11} and v_{12} . Which state variable is it beneficial to measure in order to obtain a reliable estimate of x_1 and x_2 ? Why?
- 3.3 For the choice taken in 4.2, analyze the convergence properties of the gain of the 1-step ahead Kalman predictor and write the equation of the (1-step) Kalman predictor in stationary conditions (without computing the gain).
- 3.4 Draw the block scheme of the 2-step ahead Kalman predictor.

Solution:

- 3.1 x_1 is an AR process with a pole in $z = 1$, so it is not stationary. x_2 is an ARX process (x_1 can be seen as an exogenous input) with a pole in $z = 2$, so it is not stationary.
- 3.2 It is better to measure x_2 , since measuring x_1 would lead to an insufficient knowledge of x_2 (e.g., x_2 could diverge far from x_1 and we would never know it, as x_2 does not affect x_1). More formally, if the second state is measured, the system becomes reachable and observable, thus the variance of the Kalman state estimation error converges for the second convergence theorem. If the first state is measured, the observability matrix becomes of rank 1, thus convergence cannot be guaranteed (with additional tools, divergence could be even proved).
- 3.3 The variance of the 1-step ahead prediction error converges to the solution of the ARE for what stated above. The equation of the steady-state predictor is

$$\begin{cases} \hat{x}(t+1|t) = Fx(t|t-1) + \bar{K}(y(t) - \hat{y}(t|t-1)) \\ \hat{y}(t|t-1) = H\hat{x}(t|t-1) \end{cases}$$

where $F = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$, $H = \begin{bmatrix} 0 & 1 \end{bmatrix}$ and $\bar{K}$ obtained as illustrated in Chapter 16 of the lecture notes from the solution of the ARE.

- 3.4 See Chapter 16 of the lecture notes for the block scheme of the 1-step ahead Kalman predictor. The 2-step ahead predictor is simply obtained by multiplying the 1-step ahead predictor by F .

$$\begin{aligned} F &= \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix} & H &= \begin{bmatrix} 0 & 1 \end{bmatrix} & V_1 &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow T_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ & & & & & \text{blue arrow } y = x_2(t) + v_{12}(t) & & (V_{12} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}) \\ & & & & & & & \text{blue arrow } R(1|F) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \text{At least one of the 2x2 matrices is full rank} \end{aligned}$$

4. Given the following 3rd order system:

$$\begin{cases} x_1(t+1) = \frac{1}{2}x_1(t) + \frac{3}{10}x_2(t) + g * u(t) \\ x_2(t+1) = \frac{1}{5}x_2(t) \\ x_3(t+1) = \frac{1}{2}x_3(t) - g * u(t) \\ y(t) = h * (x_1(t) + x_2(t)) \end{cases} \quad g, h \in R$$

- Discuss the observability and reachability with respect to parameters g and h .
- Apply the state transformation $\tilde{x}(t) = Tx(t)$, where $T = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ and $T^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$
- Identify the system order starting from the impulse response, considering $g = 1$ and $h = 1$.

$$F = \begin{bmatrix} \frac{1}{2} & \frac{3}{10} & 0 \\ 0 & \frac{1}{5} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix} \quad G = \begin{bmatrix} g \\ 0 \\ g \end{bmatrix} \quad H = [h \quad h \quad 0] \quad n = 3$$

$$O = \begin{bmatrix} H \\ HF \\ HF^2 \end{bmatrix} = \begin{bmatrix} h & h & 0 \\ \frac{1}{2}h & \frac{1}{2}h & 0 \\ \frac{1}{4}h & \frac{1}{4}h & 0 \end{bmatrix} \rightarrow \text{rank}(O) < n \quad \forall g, h \rightarrow \text{The system is not observable } \forall g, h$$

$$R = \begin{bmatrix} g & \frac{1}{2}g & \frac{1}{4}g \\ 0 & 0 & 0 \\ -g & -\frac{1}{2}g & -\frac{1}{4}g \end{bmatrix} \rightarrow \text{rank}(R) < n \quad \forall g, h \rightarrow \text{The system is not reachable } \forall g, h$$

$$\tilde{F} = TFT^{-1} = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{5} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix} \quad G = TG = \begin{bmatrix} g \\ 0 \\ 0 \end{bmatrix} \quad H = T^{-1}H = [h \quad 0 \quad 0] \quad \begin{cases} \tilde{x}_1(t+1) = \frac{1}{2}\tilde{x}_1(t) + gu(t) \\ \tilde{x}_2(t+1) = \frac{1}{5}\tilde{x}_2(t) \\ \tilde{x}_3(t+1) = \frac{1}{2}\tilde{x}_3(t) \\ y(t) = h\tilde{x}_1(t) \end{cases}$$

The system is composed by three independent subsystems. The input affects only the first subsystem and the output depends just on the first subsystem. Then the transfer function is (substituting the values of h and g):

$$W(z) = \frac{1}{z - \frac{1}{2}} = \frac{z^{-1}}{1 - \frac{1}{2}z^{-1}}$$

Using the geometric series:

$$\begin{aligned} W(z) &= z^{-1} \left(1 + \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2} + \frac{1}{8}z^{-3} + \frac{1}{16}z^{-4} + \frac{1}{32}z^{-5} + \dots \right) = \\ &= z^{-1} + \frac{1}{2}z^{-2} + \frac{1}{4}z^{-3} + \frac{1}{8}z^{-4} + \frac{1}{16}z^{-5} + \dots \end{aligned}$$

$$\omega(0) = 0 \quad \omega(1) = 1 \quad \omega(2) = \frac{1}{2} \quad \omega(3) = \frac{1}{4} \quad \omega(4) = \frac{1}{8} \quad \omega(5) = \frac{1}{16}$$

$$H_1 = \omega(1) = 1 \rightarrow \text{rank}(H_1) = 1 \quad H_2 = \begin{bmatrix} \omega(1) & \omega(2) \\ \omega(2) & \omega(3) \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{4} \end{bmatrix} \rightarrow \text{rank}(H_2) = 1 \rightarrow n = 1$$

5. Consider the stationary process generated as:

$$y(t) = \frac{(1-5z^{-1})(1+\frac{1}{4}z^{-1})(1-\frac{1}{4}z^{-1})}{1-\frac{1}{5}z^{-1}} e(t)$$

$e(t) \approx WN(0,1)$. Put the model in canonical form and determine the optimal predictor $\hat{y}(t+k/t)$ for $k=1, 2, 3, \dots$

Draw the diagram of the prediction error variance as a function of the prediction horizon k .

We rewrite the process equation as follows

$$y(t) = \frac{1}{(-5)} \frac{1-5z^{-1}}{1-\frac{1}{5}z^{-1}} \left(1+\frac{1}{4}z^{-1}\right) \left(1-\frac{1}{4}z^{-1}\right) (-5)e(t) = \left(1-\frac{1}{16}z^{-2}\right) \tilde{e}(t)$$

where $\tilde{e}(t) \sim WN(0,25)$. The one above is the required canonical form.

Using the long division, we compute that

$$\hat{y}(t|t-1) = \hat{y}(t|t-2) = \frac{-\frac{1}{16}}{1-\frac{1}{16}z^{-2}} y(t-2)$$

$$\hat{y}(t|t-\tau) = 0 \text{ for } \tau \geq 3$$

Correspondingly, denoting $\varepsilon(t|t-\tau) = y(t) - \hat{y}(t|t-\tau)$, we compute that

$$\varepsilon(t|t-1) = \varepsilon(t|t-2) = \tilde{e}(t)$$

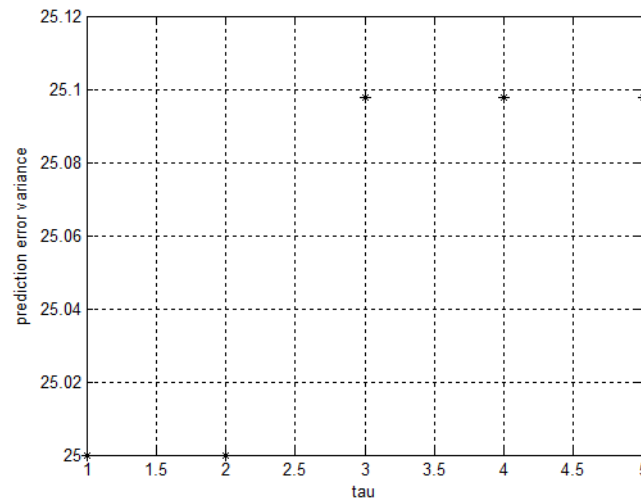
$$\varepsilon(t|t-\tau) = y(t) \text{ for } \tau \geq 3$$

Therefore

$$\text{var}(\varepsilon(t|t-1)) = \text{var}(\varepsilon(t|t-2)) = 25$$

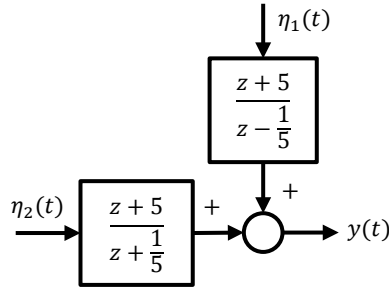
$$\text{var}(\varepsilon(t|t-\tau)) = \left(1 + \left(\frac{1}{16}\right)^2\right) \cdot 25 \text{ for } \tau \geq 3$$

The corresponding plot is



6. Consider the process $y(t)$ at the output of the system in the figure below:

where $\eta_1(t) \sim WN(1,1)$, $\eta_2(t) \sim WN(0,1)$ are uncorrelated, namely



$$E[(\eta_1(t_1) - E[\eta_1(t_1)])(\eta_2(t_2) - E[\eta_2(t_2)])] = 0, \forall t_1, t_2$$

Is $y(t)$ a stationary process? Why?

Compute the mean value of $y(t)$.

Find the expression of the spectrum $\Gamma_y(\omega)$ of $y(t)$.

Draw the qualitative plot of the spectrum.

The process is stationary because it is the sum of two stationary processes, i.e., $y_1(t)$ and $y_2(t)$, where

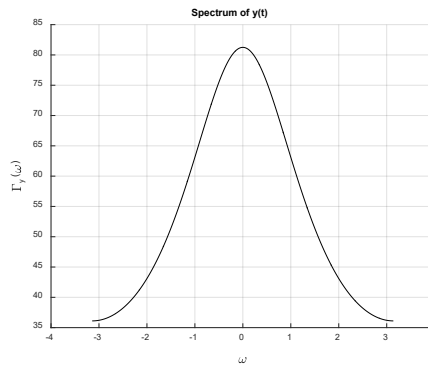
$$y_1(t) = \frac{z+5}{z-\frac{1}{5}} \eta_1(t), y_2(t) = \frac{z+5}{z+\frac{1}{5}} \eta_2(t)$$

The processes $y_1(t)$ and $y_2(t)$ are stationary, since they are generated as filtered white noises, with stable filters (with poles that lie strictly inside the unit circle). We compute that

$$\bar{y} = E[y(t)] = E[y_1(t)] + E[y_2(t)] = \frac{15}{2} + 0$$

Since $\eta_1(t)$ and $\eta_2(t)$ are uncorrelated, also $y_1(t)$ and $y_2(t)$ are uncorrelated with each other. In view of this, $\Gamma_y(\omega) = \Gamma_{y_1}(\omega) + \Gamma_{y_2}(\omega)$, where $\Gamma_{y_1}(\omega) = \frac{26+10\cos(\omega)}{25-\frac{2}{5}\cos(\omega)}$ and $\Gamma_{y_2}(\omega) = 25$, in view of the fact that $y_2(t)$ is indeed a white noise process. We obtain that

$\Gamma_y(\omega) = \frac{52}{\frac{26}{25}-\frac{2}{5}\cos(\omega)}$. The plot of spectrum of $y(t)$ is depicted below.



7. Consider the state-space (SS) representation of a discrete-time Linear Time Invariant system. Starting from this representation:

- Find the formula to obtain the Transfer-Function representation
- Find the formula to obtain the Impulse-Response representation

See notes of the corresponding chapter